

Tanishq Pauskar

330-904-5184 | tpauskar@purdue.edu

EDUCATION

Purdue University, West Lafayette, Indiana

B.S. in Computer Science and B.S. in Mathematics (GPA: 3.90)

Aug 2024 – May 2028

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java, JavaScript, SQL, R, HTML/CSS, LaTeX
- **Frameworks & Tools:** React, Node.js, Tailwind CSS, PostgreSQL, Firebase
- **AI/ML:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, Multi-Agent System

AWARDS

Putnam Top 200 (2024-2025): Ranked 109 of 3,988 in North America's most prestigious collegiate math competition.

USA Math Olympiad Bronze Medalist (2024): Ranked 44th nationally in USA's premier math competition.

USA Physics Olympiad Bronze Medalist (2023): Top 200 rank in USA's most prestigious high school physics competition.

Competitive Programming: **Codeforces Candidate Master** (Rating: 2079, top 2.5% globally)

WORK EXPERIENCE

Undergraduate Security Researcher

May 2026 – Present

Trustworthy Security Laboratory, Purdue University, West Lafayette, Indiana

- Discovered a novel black-box membership inference attack for diffusion models, improving detection by 94% at 1% false-positive rate.
- Built an end-to-end black-box LiRA framework and validated the signal on CIFAR-10 and STL-10 datasets across DDPM and fine-tuned Stable Diffusion target models.

Teaching Assistant

Jan 2026 – Present

Purdue University Computer Science Department, West Lafayette, Indiana

- Led office hours and weekly study sessions for Discrete Mathematics (Spring 2026) and Competitive Programming I (Fall 2026).
- Graded assignments, proctored exams, and assisted students with course material and problem solving.

Undergraduate Security Researcher

May 2025 – Dec 2025

Trustworthy Security Laboratory, Purdue University, West Lafayette, Indiana

- Demonstrated privacy vulnerabilities in GIS-rendered maps in medical publications; Co-authored paper accepted to **ACM CCS '26 conference**.
- Built a Python image processing pipeline in PIL to detect map marker locations accurately. Developed algorithms combining gradient descent optimization, geometric modeling, and K-means clustering to recover sub-pixel coordinates within 2 meters on continental-scale maps.

PROJECTS

Purdue Math Club POTW | ReactJS, Node.js, PostgreSQL

May 2026 – Present

- Built and deployed a React/Express platform for Purdue Math Club's Problem of the Week, used by 50+ users to submit LaTeX/PDF solutions, propose problems, and request hints.
- Designed a Render-hosted PostgreSQL backend to archive past POTWs, store solution submissions, and manage user profiles, leaderboards, feedback, problem proposals, and JWT-based authentication.

POSIX-Style UNIX Shell | POSIX, C, C++, Lex, Yacc

Jan 2026 – Apr 2026

- Engineered a Unix-compatible shell in C using Lex/Yacc for parsing. Implemented process management, pipelines, I/O redirection, and background job execution using fork, execvp, and dup2 system calls.
- Deployed advanced Unix shell features, including job control with signal handling (SIGINT, SIGTSTP), subshell execution, wildcard expansion, and a custom line editor with command history.

Connect 4 Human-Replicable Solver | C++, ReactJS, Node.js, PostgreSQL, Socket.io

June 2025 – Aug 2025

- Developed a novel lightweight perfect-play strategy for Connect 4 by solving its ~4 trillion-state search space with minimax and alpha-beta pruning, then compressing the solution into a ~10K-node opening book.
- Built a real-time multiplayer Connect 4 application and perfect-play trainer using React and Socket.io, with cross-device game synchronization and matchmaking. Designed a PostgreSQL database on Supabase to store user profiles and game histories and support dynamic Elo ratings and real-time leaderboards.

Tweets Classification using Machine Learning | NLP, Keras, Python, TensorFlow

Oct 2025 – Dec 2025

- Developed a multilayer perceptron (MLP) to classify disaster-related tweets, achieving 85%+ accuracy to help first responders prioritize critical information; trained and evaluated the model in TensorFlow using TF-IDF text embeddings.

EXTRACURRICULARS

Vice President of Purdue Math Club: Organized new programs and events that helped more than double club attendance and participation in club events. Conducted weekly lectures to help students prepare for Putnam Math Contest.

Apr 2025 – Present

Competitive Programming Team: Represented Purdue University's team #2 at ICPC contest.

Oct 2025 – Present